FIITJEE

# JEE (Main)-2018

ALL INDIA TEST SERIES

## OPEN TEST

**Time Allotted: 3 Hours** **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

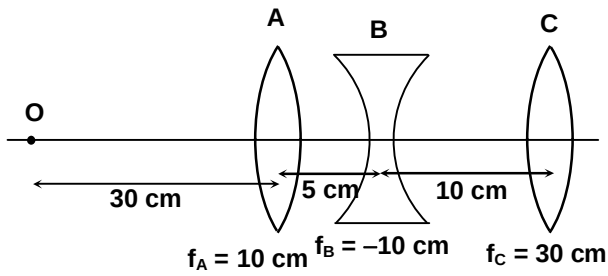
# Physics

**PART – I**

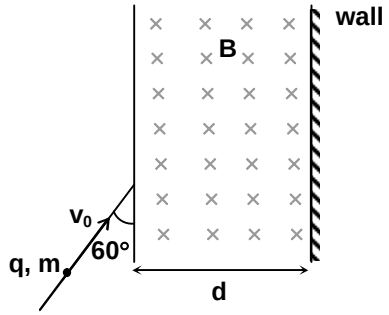
## SECTION – A
**(One Options Correct Type)**

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A point object 'O' lies in front of the lens as shown in figure. Find the final position of image formed by the combined lenses.
(A) 10 cm to the right of the lens C
(B) 15 cm to the right of the lens C
(C) 20 cm to the right of the lens C
(D) 30 cm to the right of the lens C




2. A positively charged particle (q, m) enters in a uniform magnetic field 'B' at an angle of 60° as shown in figure with speed $v_0$. Collision between the charge particle and the wall is perfectly elastic. Then find the time in which the charge particle comes out from the magnetic field. $\left(\text{take } d = \frac{(\sqrt{3}-1)mv}{2qB}\right)$

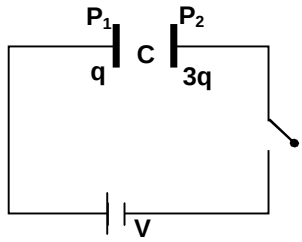



(A) $\frac{\pi m}{2qB}$ (B) $\frac{\pi m}{6qB}$

(C) $\frac{\pi m}{3qB}$ (D) $\frac{2\pi m}{3qB}$

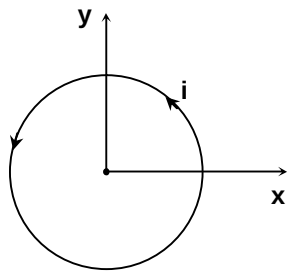
***Space for Rough work***

3. A capacitor of capacity C having charge q and 3q on the plate $P_1$ and $P_2$ respectively as shown in figure. Now switch 'S' is closed, find the work done by the battery. (take q = CV)




(A) qV (B) 2qV
(C) 3qV (D) 4qV

4. Magnetic field in a space is given as $\vec{B} = -\frac{B_0 x}{R}\hat{k}$. A circular ring of radius R having current I lies in the space with centre at origin as shown in the figure. Find the magnitude of net magnetic force acting on the circular ring due to the magnetic field.

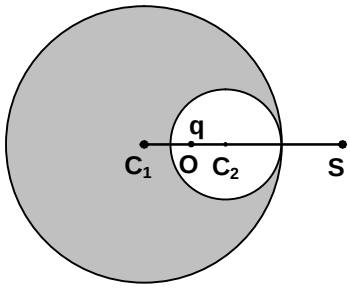



(A) $2iB_0R$ (B) $\pi iB_0R$
(C) $2\pi iB_0R$ (D) zero

5. A charge particle is moving with a velocity $3\hat{i} + 4\hat{j}$ m/sec and it has electric field $\vec{E} = 10\hat{k}$ N/C at a given point. Find the magnitude of magnetic field at the same point due to the motion of the charge particle.

(A) 100 $\mu_0\varepsilon_0$ (B) 25 $\mu_0\varepsilon_0$
(C) 12.5 $\mu_0\varepsilon_0$ (D) 50 $\mu_0\varepsilon_0$

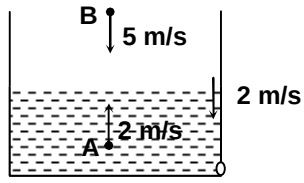
6. A charge q is placed in the cavity at 'O' in a spherical uncharged conducting conductor. Point 'S' is outside the conductor. If charge 'q' is displaced towards $C_2$. Then




(A) Electric field due to induced charge at S increases
(B) Electric field due to induced charge at S decreases
(C) Electric field due to induced charge at S first increase then decrease.
(D) Electric field due to induced charge at S will not change.

***Space for Rough work***

7. Two point objects are moving towards each other. Object 'A' is inside the waver ($\mu_w$ = 4/3) and object B is in the air. Water level is moving downward with a velocity 2 m/s as shown in figure. Find the speed of A with respect to B.



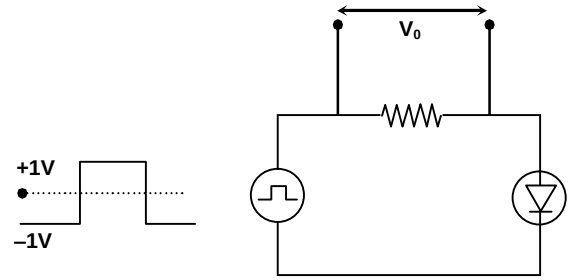

(A) 5 m/s (B) 7 m/s
(C) 8 m/s (D) 6 m/s

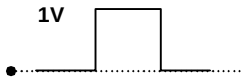
8. Light of wavelength 488 nm is produced by an Argon laser which is used in the photoelectric effect. When light from this spectral line is incident on the emitter, the stopping (cut-off) potential of photo electrons is 0.38 V. Find the work function of the material form which the emitter is made.
(A) 1.38 eV (B) 2.55 eV
(C) 2.17 eV (D) 2.93 eV

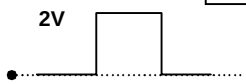
9. When an electric field is applied across a semiconductor
(A) Electrons move from higher energy level to the lower energy level in the conduction band.
(B) Electrons in conduction band remains in rest
(C) holes in the valance band move from higher energy level to lower energy level.
(D) holes in the valance band move from lower energy level to higher energy level.

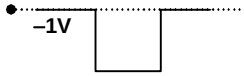
10. In a n-p-n transistor circuit, the collector current is 10 mA. If 80 percent of electrons emitted reach to the collector. Then which of the following statement is true.
(A) The emitter current will be 2.5 mA
(B) The emitter current will be 22.5 mA.
(C) The base current will be 2.5 mA.
(D) The base current will be 22.5 mA

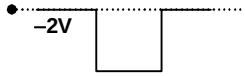
11. The output waveform across the resistor is




(A) 

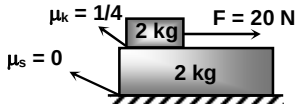

(B) 


(C) 


(D) 

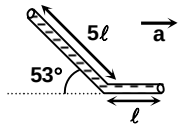

***Space for Rough work***

12. A force of 20 N is applied on upper block as shown in the figure. The total work done by frictional during the time interval in which the upper block has a displacement of 15 m with respect to the ground is (take g = 10 m/s$^2$)




(A) 50 J (B) $-50$ J
(C) 75 J (D) $-75$ J

13. In the given tube liquid density $\rho$ is filled. Find the acceleration of the tube towards right direction such that no liquid will fall out of the tube.




(A) $\frac{3}{5}g$ (B) $\frac{4}{5}g$
(C) g (D) 4g

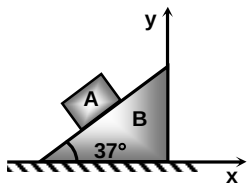
14. An artificial satellite is moving in a circular orbit around the earth with a speed equal to 3/8 times of the magnitude of escape velocity from the earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely onto the earth, then the speed with which it hits the surface of the earth is (take g = 10 m/s and $R_e$ = 6400 Km)

(A) $\sqrt{\frac{29}{4}}$ Km/s (B) $\sqrt{\frac{23}{2}}$ Km/s
(C) $\sqrt{\frac{29}{6}}$ Km/s (D) $\sqrt{\frac{29}{13}}$ Km/s

15. An unknown resistance r is determined in terms of a standard resistance R = 100 $\Omega$ by using potentiometer. The potential difference across r is balanced against 45 cm length of the wire and that (r + R) is obtained at 70 cm of the wire. Find the value of the unknown resistance.

(A) 200 $\Omega$ (B) 280 $\Omega$
(C) 180 $\Omega$ (D) 100 $\Omega$

16. In the figure shown the velocity of A with respect to ground is $(20\hat{i}+15\hat{j})$ m/s, then what will be the magnitude of velocity of the wedge 'B' with respect to ground frame. (A remains in contact with B)




(A) 12 m/s (B) 25 m/s
(C) 20 m/s (D) zero

***Space for Rough work***

17. An object is subjected to an acceleration a = 4 + 3v. It is given that the displacement S = 0 when v = 0. The value of displacement when v = 3 m/s is

(A) $1+\frac{4}{3}\ell n\left(\frac{13}{4}\right)$ (B) $1-\frac{4}{3}\ell n\left(\frac{13}{8}\right)$

(C) $1-\frac{4}{9}\ell n\left(\frac{13}{4}\right)$ (D) $1+\frac{4}{9}\ell n\left(\frac{13}{8}\right)$

18. A progressive wave travels in a medium $M_1$ and enters into another medium $M_2$ in which its speed decreases to 60%. Then the ratio of the amplitude of the transmitted and the incident waves is

(A) 3/5 (B) 3/4

(C) 3/10 (D) 1

19. A wave is travelling along x-axis. The disturbance at x = 0, and t = 0 is A/2 and is increasing, where A is amplitude of the wave. If Y = A sin (kx – ωt + ϕ), then the initial phase is

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$

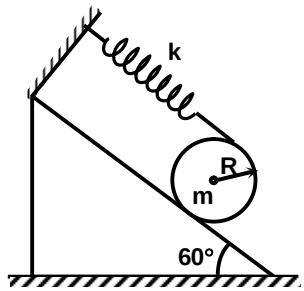
(C) $\frac{\pi}{3}$ (D) $\frac{11\pi}{6}$

20. A block of mass 'm' is attached by an inelastic string to a suspended spring of spring constant 'k'. Both the string and the spring have negligible mass and the string inextensible and of length 'L'. Initially, the block is at rest. The largest amplitude $A_{max}$, for which the string will remain taught in the motion.

(A) $\frac{mg}{2k}$ (B) $\frac{mg}{k}$

(C) L/2 (D) L

21. A uniform solid sphere of mass 'm' and radius R is in equilibrium on an inclined plane by the action of the light spring of stiffness k, gravity and on it. If the angle of inclination of the plane is 60°, then find the time period of oscillation for small displacement.




(A) $2\pi\sqrt{\frac{m}{10k}}$ (B) $2\pi\sqrt{\frac{7m}{10k}}$

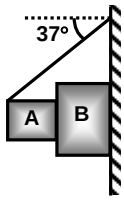
(C) $2\pi\sqrt{\frac{7m}{20k}}$ (D) $2\pi\sqrt{\frac{7m}{40k}}$

***Space for Rough work***

22. Two identical containers A and B having same volume of an ideal gas at same temperature have mass of the gas as $m_1$ and $m_2$ respectively and $2m_1 = 3m_2$. The gas in each cylinder expands isothermally to double of its volume. If change in pressure in A is 300 Pa, then the change in pressure in B is
(A) 200 Pa (B) 300 Pa
(C) 400 Pa (D) 500 Pa

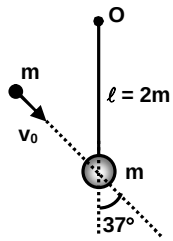
23. The mass of a gas molecule can be computed from the specific heat at constant volume. $C_V$ for argon is 0.15 Kcal/kg-k. The molecular weight of an argon atom is (approximately) (R = 2 cal/mol-K)
(A) $5 \times 10^{-3}$ kg (B) $2.5 \times 10^{-3}$ kg
(C) $40 \times 10^{-3}$ kg (D) $20 \times 10^{-3}$ kg

24. Two block A and B of mass 50 kg and 300 kg respectively are placed as shown in figure. Coefficient of friction between all surface is $\frac{1}{2}$.




Then (take g = 10 m/s)
(A) Tension in the string is 2000 N
(B) Tension in the string is 3000 N
(C) Acceleration of block B is 5/3 m/s$^2$.
(D) Acceleration of block B is 10/3 m/s$^2$.

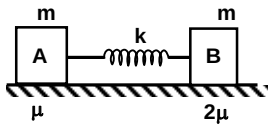
25. An elastic ball of mass 'm' is suspended from a fixed point 'O' by an inextensible string of length 2m. A small particle of mass m moving downward at angle of 37° with the vertical hits the ball with $v_0$. The coefficient of restitution for collision is 4/5. the velocity of the particle $v_0$ such that ball just complete one revolution after the collision.




(A) $\frac{170}{27}$ m/s (B) $\frac{85}{27}$ m/s
(C) $\frac{680}{27}$ m/s (D) $\frac{340}{27}$ m/s

***Space for Rough work***

26. Two block of same mass 'm' are joined with an ideal spring of spring constant k and kept on a rough surface as shown in figure. The spring is initially unstretched and coefficient of friction between the blocks and horizontal surface is $\mu$ and $2\mu$ respectively. What should be the maximum speed of block 'A' such that the block 'B' does not move?




(A) $2\mu g\sqrt{\frac{2m}{k}}$ (B) $\mu g\sqrt{\frac{2m}{k}}$

(C) $\mu g\sqrt{\frac{m}{k}}$ (D) $2\mu g\sqrt{\frac{m}{2k}}$

27. A block weight is 15 Newton in air. It weight 12 Newton when immersed in water. When immersed in another liquid, it weight is 13 Newton. Then find the ratio of density of block and the other liquid. ($\rho_{block}/\rho_{liquid}$)

(A) $\frac{15}{2}$ (B) 15

(C) $\frac{15}{4}$ (D) $\frac{13}{12}$

28. A bar magnet having a magnetic moment of $1.0 \times 10^4$ J/T is free to rotate in a horizontal plane. A horizontal magnetic field $B = 4 \times 10^{-5}$ T exists in the space. The work done in rotating the magnet slowly from a direction parallel to a direction 60° from the field will be:

(A) 0.2 J (B) 0.1 J

(C) 0.4 J (D) 0.3 J

29. A person is viewing two red dots at a distance of 1000 m. What should be the minimum distance between dots so that his eye can resolve two dots separately.
(take : $\lambda_{red}$ = 700 nm, pupil dia = 2.5 mm)

(A) 45 cm (B) 50 cm

(C) 35 cm (D) 10 cm

30. A parallel beam of light of wavelength $\lambda$ passes through a slit of width d. The transmitted light is collected on a screen D away (D >> d). Find the distance between the two second order minima.

(A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$

(C) $\frac{2D\lambda}{d}$ (D) $\frac{4D\lambda}{d}$

***Space for Rough work***

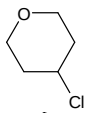
# Chemistry

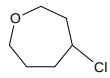
**PART – II**

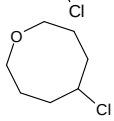
**SECTION – A**
**Straight Objective Type**

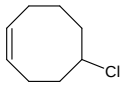
This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. Which of the following undergoes fastest solvolysis?

(A) 


(B) 


(C) 


(D) 

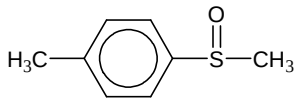

32. Which of the following is correct for various reversible reactions?

(A) In case of liq. $\rightleftharpoons$ gas; systems the equilibrium is attained between both the phases, whether the system is closed or open.
(B) Boiling point of a liquid is independent of the altitude of the place of experiment.
(C) The value of equilibrium constant is independent of concentration terms.
(D) If a reversible reaction has high value of its equilibrium constant, the reaction quickly attains the equilibrium.

33. Which of the following is chiral as well optically active?

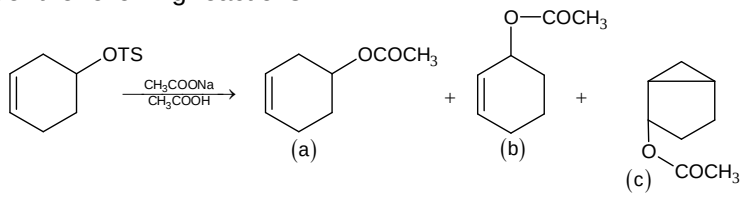
(A) Me – NH-Et

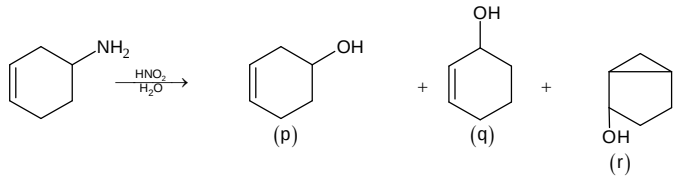
(B) cis-cyclooctene

(C) 


(D) $Et.\dot{C}HMe$

34. Consider the following reactions:

(I) 

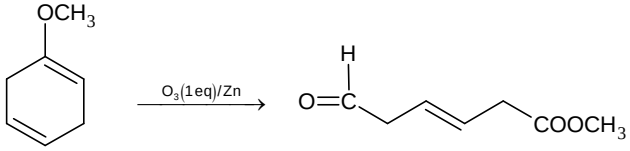

(II) 

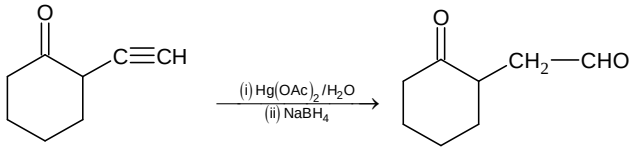

Which of the following is correct combination of major products, correspondingly?

(A) I – a, II – p
(B) I – a, II – q
(C) I – b, II – q
(D) I – b, II – p

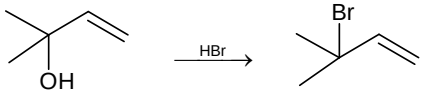
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35. Consider the following reactions:

(I) 


(II) 


(III) $Ph-CH_2-CH_2-\overset{\oplus}{N}(Et)_3 \xrightarrow[\Delta]{\text{Moist } Ag_2O} PhCH=CH_2+Et_3N$

(IV) 

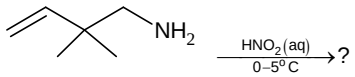

In which of the above reactions, the products written are major products.

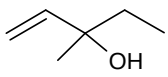
(A) I, II, III & IV
(B) I, III & IV only
(C) I, II & III only
(D) II & IV only

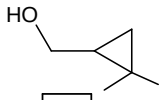
36. $BrF_3$ and $IF_5$ both exist in ionic form, in solid state. What are the hybridizations of cationic parts of both, respectively?

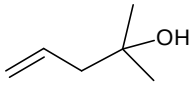
(A) $sp^3$, $sp^3$
(B) $sp^3d$, $sp^3d$
(C) $sp^3$, $sp^3d$
(D) $sp^2$, $sp^3$

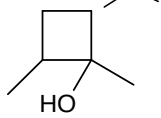
37. What is the major product of the following process?




(A) 

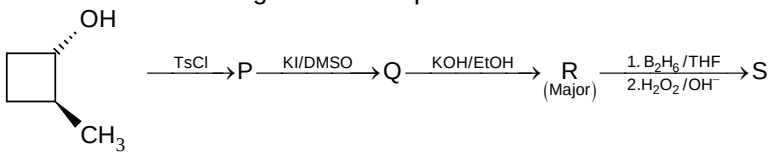

(B) 


(C) 

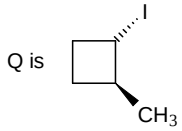

(D) 

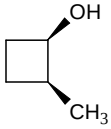

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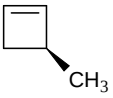
38. Consider the following reaction sequence:

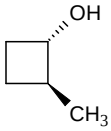



Choose the most correct of the following?

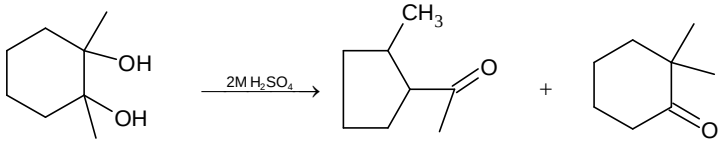
(A) Q is 


(B) S is 

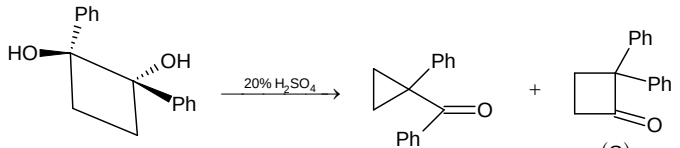

(C) R is 


(D) S is 

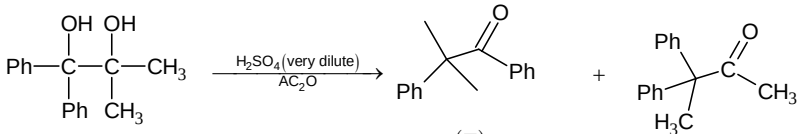

39. Consider the following reactions:

(I) 


(P) (Q)

(II) 


(R) (S)

(III) 

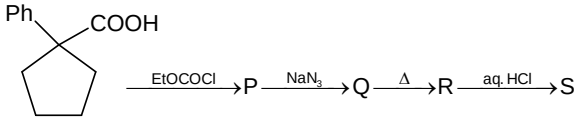

(T) (U)

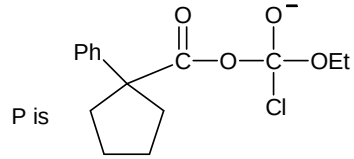
Select the correct statement from the following:
(A) P is more than Q.
(B) S is not formed in the reaction (II).
(C) T is almost 100% product of the reaction (III).
(D) All others are also correct.

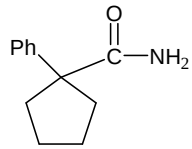
40. Glucose and fructose can be distinguished by:
(A) Fehling solution
(B) Osazone formation
(C) Bromine water
(D) Conc. $H_2SO_4$/warming.

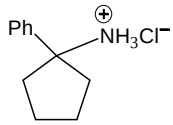
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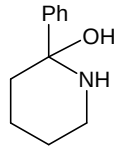
41. Consider the following reaction sequence and select the correct one from the given statements?




(A) P is 

(B) Q is 

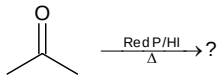
(C) S is 

(D) S is 

42. Which of the following reactions end with the same product?

(I) $(CH_3)_2C=N-OH \xrightarrow[\Delta]{OH^-} ?$

(II) $(CH_3)_2C=N.NHSO_2Ph \xrightarrow{NaBH_4} ?$

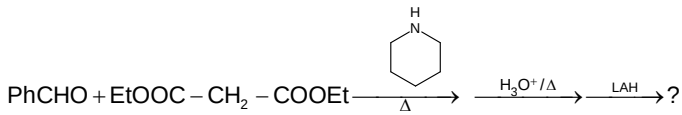
(III) 

(IV) $CH_3-\underset{OH}{\underset{|}{C}}HCHO + NH_2OH \longrightarrow \xrightarrow{AcOH/AC_2O} \xrightarrow{\text{Moist } Ag_2O} \xrightarrow{\text{Red P/HI}} ?$

(A) I, III
(B) I, II, III, IV
(C) I, II, III
(D) II and III only

43. Which of the following reactions end with the same product?

(a) $PhCH=CHCH_2OH \xrightarrow{MnO_2} ?$

(b) 

(c) $PhCHO \xrightarrow[\Delta(5-6\ hr)]{AC_2O/ACO^-} \xrightarrow{LAH} \xrightarrow{MnO_2} ?$

(A) a, b, c
(B) a, c
(C) b, c
(D) a, b

44. 7.5 gm glycine is added to 500 ml water. What is the pH of the solution? (Given : $pK_{a_1}(\text{glycine}) = 2.4; pK_{a_2}(\text{glycine}) = 9.54$)

(A) 6.15
(B) 2.58
(C) 9.58
(D) 5.97

***Space for Rough work***

45. The cell reaction for the given cell is spontaneous if :
$Pt;H_2(P_1\ atm)|H^+(1M)|H_2(P_2\ atm);Pt$
(A) $P_1 > P_2$ (B) $P_1 < P_2$
(C) $P_1 = P_2$ (D) $P_2 = 1$ atm

46. NaCl and NaCN can be distinguished by using the following reagents in the given sequence?
(A) (i) $H_2O_2 / \Delta$ (ii) $AgNO_3/HNO_3$ (B) $HgNO_3(aq)$
(C) (i) Conc. $H_2SO_4$ (ii) combustion of gas evolved (D) By all of A, B & C

47. Calculate the equilibrium constant for the following reaction:
$3Sn + 2Cr_2O_7^{2-} + 28H^+ \longrightarrow 3Sn^{4+} + 4Cr^{3+} + 14H_2O$
$E^o_{Sn/Sn^{4+}} = 0.009$ V; $E^o_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33$ V
(A) $10^{413}$ (B) $10^{1.8}$
(C) $10^{272}$ (D) $10^{317}$

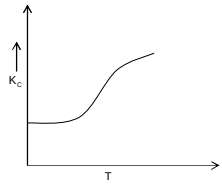
48. Which of the following is incorrect?
(A) For a perfect gas : at constant temperature, the product of pressure and specific volume $\left(\frac{V}{M}\right)$ is constant.
(B) The zero of absolute temperature is that temperature at which a perfect gas of finite volume has no pressure.
(C) On the basis of second law of thermodynamics, a definition of temperature completely independent of the properties of any individual substance and applicable to all stages of heat and cold.
(D) Other three are incorrect.

49. Which of the following statements is correct?
(A) Heat capacity of solids and liquids is almost independent of any variations of external pressure, that may take place during the process of heating.
(B) Dulong and Petits law (at. Wt. × Sp. Heat = 6.4) is better applicable to the substance/element of lower atomic/molecular mass.
(C) Applicability of Dulong & Petits law to molecular species is independent of molecular structures/constitutions.
(D) Specific heat of a gas is not influenced considerably by the conditions of the heating process.

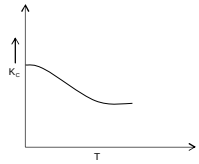
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50. Which of the following is correct?

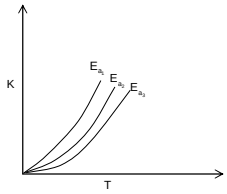
(A) For a reversible reaction for which $\Delta_r H = -ve$




(B) For reversible reaction for which $\Delta_r H = +ve$




(C)




;

$E_a$ = Activation energy.
$E_{a_1} > E_{a_2} > E_{a_3}$;
K = Specific rate constant

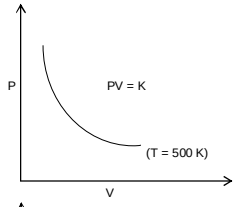
(D) Activation energy of a reaction is a barrier to energy transfer (KE to PE) between reacting molecules that must be overcome, which is due to need of bonds stretch as well as steric repulsions.

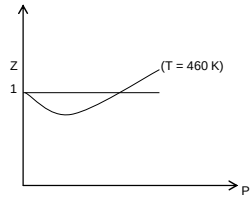
51. An ideal equimolar binary liquid solution of A and B has a vapour pressure of 900 mm of Hg. It is distilled at a constant temperature till $\frac{2}{3}$ $(\text{i.e.}66.67\%)$ of the original amount. If the mole fraction of A is 0.3 in residue and that of B in condensate is 0.4 and also the vapour pressure of residue is 860 mm Hg which of the following is correct?

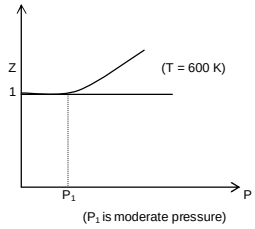
(A) Mole fraction of A is more than that of B in vapour phase, above residue as well as above condensate.
(B) Vapour pressure of condensate is more than 900 mm Hg at same temperature.
(C) In vapour pressure of residue, vapours of A are more contributing.
(D) If the solution would be distilled till 50% of the original amount then mole fraction of B in liquid condensate would be more than that of A in same solution.

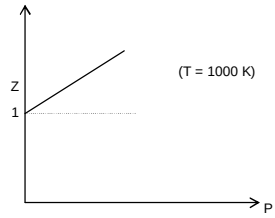
***Space for Rough work***

52. If the critical temperature of a gas is 500 K, then which of the following is correct for the same gas?

(A) 


(B) 


(C) 


(D) 


53. Consider the following reactions:

$Fe^{2+} \xrightarrow{\text{excess } CN^-} P$;

$Fe^{3+} \xrightarrow{\text{excess } CN^-} Q$;

Which of the following is correct about P and Q?

(A) $P + Fe^{2+} \longrightarrow$ Prussian blue coloured precipitate.

(B) Iron in both the P and Q, is $d^2sp^3$ hybridised.

(C) Complex P is paramagnetic.

(D) Complex 'Q' has smaller magnetic moment than P.

54. Which of the following order is correct regarding dipole moment?

(A) $CO_2 < SO_2 < SO_3$ (B) $PCl_5 < PCl_3F_2 < PCl_2F_3$

(C) $XeF_6 > XeF_4 = XeF_2$ (D) $CH_4 < CH_2Cl_2 < CH_3Cl < CHCl_3$

55. In a fcc lattice A, B, C and D atoms are arranged at corners, face centres octahedral voids and half of tetrahedral voids respectively and if the crystal deposited is defected such that all the particles at one body diagonal of each unit cell are missing, correspondingly then what is the resulted formula of the compound?

(A) $AB_3C_4D_4$ (B) $AB_4C_4D_4$

(C) $A_3B_3C_2D_4$ (D) $A_2B_3C_3D_4$

56. Which of the following lanthanoid ions, has zero magnetic moment? [Ce (z = 58); Sm (z = 62); Eu (z = 63); Yb (z = 70)]

(A) $Sm^{2+}$ (B) $Eu^{2+}$

(C) $Yb^{2+}$ (D) $Ce^{2+}$

***Space for Rough work***

57. Which of the following is correct?
(A) All alkali earth metal salts impart a characteristic colour to Bunsen flame.
(B) CsOH is most soluble hydroxide among LiOH, KOH, RbOH and CsOH.
(C) Among the alkali metals Li is most reactive under all the conditions.
(D) The reducing character of the alkali metal hydrides follow the following order:
LiH > NaH > KH > RbH > CsH

58. In the metallurgical process of iron, when lime stone is added to the blast furnace, the $Ca^{2+}$ ions end-up in:
(A) Metallic calcium
(B) Gangue
(C) $CaSO_3$
(D) Slag

59. Which of the following cannot decolourise the iodine solution?
(A) Excess of $Cl_2$ water.
(B) Excess of $Na_2S_2O_3$ solution.
(C) Excess of $Br^-$ solution.
(D) Excess of NaOH solution.

60. (i) Inorganic compound $'A'(aq) + \text{DeVarda's alloy} \xrightarrow{\Delta} B(g) + \text{salt}$
(ii) $'A'(aq.) + C \xrightarrow{\Delta} PH_3(g)$;
(iii) $A(aq) + NH_4Cl \xrightarrow{\Delta} D(g) + Cl^- + H_2O$;
(iv) B is colourless combustible lightest gas.
(v) D-turns red litmus blue.
Which of the following combination is correct?
(A) A = KOH
B = $H_2$
C = $H_3PO_3$
D = $NH_3$
(B) A = NaOH
B = $H_2$
C = $H_3PO_4$
D = $PH_3$
(C) A = NaOH
B = $H_2$
C = $H_3PO_4$
D = $NH_3$
(D) A = NaOH
B = $H_2$
C = $P_4$
D = $NH_3$

***Space for Rough work***

# Mathematics

**PART – III**

**SECTION – A**
**Straight Objective Type**

This section contains **30 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

61. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $|\vec{c}| = 9$ & $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = -41$, then $|\vec{a}\times\vec{b}|$ is equal to

(A) 0 (B) 1
(C) 2 (D) none of these

62. The curve represented by the equation $\frac{x^2}{\sin\sqrt{2} - \sin\sqrt{3}} + \frac{y^2}{\cos\sqrt{2} - \cos\sqrt{3}} = 1$ is

(A) an ellipse with focus on x-axis (B) a hyperbola with focus on x-axis
(C) an ellipse with focus on y-axis (D) a hyperbola with focus on y-axis

63. Let $\alpha = e^{\frac{i9\pi}{19}}$, then $\text{Re}\left(\sum_{r=1}^{9}(-1)^{r+1}\cdot\alpha^r\right)$ is equal to

(A) $-\frac{1}{2}$ (B) 0
(C) $\frac{1}{2}$ (D) none of these

64. Sum of $\left(2 + \frac{10}{3} + \frac{20}{3} + \frac{68}{5} + \frac{416}{15} + \ldots\ldots \text{upto } 31 \text{ terms}\right)$ is

(A) $2^{27}\cdot 27$ (B) $2^{31}\cdot 31$
(C) $2^{27}\cdot 31$ (D) $2^{31}\cdot 27$

65. In a triangle ABC, A ≡ (1, 3) and perpendicular bisector of AB is y = x and orthocentre of triangle ABC is (3, 1), then locus of vertex C is

(A) y = x + 1 (B) y = x – 2
(C) 2y = x + 4 (D) 2y = x

***Space for rough work***

66. Let roots of the equation $x^4 - 4x^3 + 6x^2 - 4x - 15 = 0$ are $\alpha, \beta, \gamma$ and $\delta$, then the value of $\frac{\alpha-1}{\beta-1}+\frac{\beta-1}{\gamma-1}+\frac{\gamma-1}{\delta-1}+\frac{\delta-1}{\alpha-1}$ is equal to
(A) –4 (B) –2
(C) 0 (D) 4

67. Let shortest distance between a variable line and $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ lies along the points A(3, 5, 7) and B(2, 3, 4), then locus of the points on the variable line is
(A) 3x + 2y + z = 16 (B) x + 2y + 3z = 20
(C) 2x + y + 3z = 19 (D) x + 2y + 3z = 34

68. Let f(x) and g(x) be two functions such that $g(x)=\frac{[x]f(x)}{[x]f(x)}$ and g(x) is periodic function with fundamental period 'T', then minimum value of 'T' is (where [.] denotes the greatest integer function)
(A) $\frac{1}{2}$ (B) 1
(C) $\frac{3}{2}$ (D) none of these

69. Let $f(x)=\int_0^4 |x^2-4x-t|\,dt$, $x \in (0, 4)$, then the range of f(x) is
(A) (8, 24) (B) [8, 24)
(C) (8, 24] (D) [0, 8)

70. Let 'r' be the radius of a circle which is inscribed in the region made by $y^2 = 4x$, $y = x + 1$ and $y = 0$, the [r] is (where [.] denotes the greatest integer function)
(A) 1 (B) 2
(C) 3 (D) none of these

71. Let there are 4 red, 3 yellow and 2 green balls, then total number of arrangements in a row such that no two balls of same colour are together (balls of same colour are identical) are
(A) 59 (B) 79
(C) 99 (D) none of these

***Space for rough work***

72. Let A(1, 0), B(3, 4) and C(–1, 1) are the vertices of a triangle ABC and circles are drawn taking AB, BC and CA as diameters. If 'D' is the radical centre of these circles then distance of 'D' from origin is
(A) 0 (B) 1
(C) 2 (D) 3

73. A flagstaff 5m high is placed on a building 25 m high. If the flag and building both subtend equal angles on the observer at a height 30 m, then the distance between observer and the top of flag is
(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{\frac{2}{3}}$
(C) $5\sqrt{\frac{3}{2}}$ (D) $\frac{5\sqrt{2}}{3}$

74. Let p and q be two statements. Then $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
(A) tautology (B) contradiction
(C) neither tautology non contradiction (D) none of these

75. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The coefficient of variation is
(A) 10% (B) 40%
(C) 50% (D) none of these

76. Let $a + b = 3 - \cos 4\theta$ and $a - b = 4 \sin 2\theta$, then maximum value of 'ab' is equal to
(A) $\frac{1}{2}$ (B) 1
(C) $\frac{2}{3}$ (D) 3

77. In a triangle ABC, $\angle ABC = 45^\circ$ and a point D is on BC such that 2BD = CD and $\angle DAB = 15^\circ$, then $\angle ACB$ is equal to
(A) $15^\circ$ (B) $30^\circ$
(C) $60^\circ$ (D) $75^\circ$

78. A butterfly randomly lands on a face of a dice numbered from 1 to 6 and then moves to the adjacent face, then the probability that the butterfly ends up on face numbered 2 is
(A) $\frac{1}{6}$ (B) $\frac{1}{3}$
(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

***Space for rough work***

79. In an examination consisting of four parts A, B, C and D, each part containing 25 questions. If a student randomly attempts the exam, then the probability that he attempts atleast 50 questions in the exam is

(A) $\frac{1}{2}+\frac{^{100}C_{50}}{2^{101}}$ (B) $\frac{1}{2}-\frac{^{100}C_{50}}{2^{101}}$

(C) $\frac{1}{2}+\frac{^{100}C_{50}}{2^{100}}$ (D) $\frac{1}{2}-\frac{^{100}C_{50}}{2^{100}}$

80. Let a circle taking foci of the hyperbola $3x^2 + by^2 + 8xy - 10x + 1 = 0$ as end points of diameter, meet the asymptotes of the hyperbola at A, B, C and D, then $\frac{2b^2}{3}$ is equal to (ABCD is a square)

(A) 1 (B) 2

(C) 3 (D) none of these

81. From a point P(–4, 9) tangents $PT_1$ and $PT_2$ are drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at points $T_1$ and $T_2$ respectively. If $F_1\left(-\sqrt{7},0\right)$ and $F_2\left(\sqrt{7},0\right)$ be two points and $S_1$ & $S_2$ be the images of $F_1$ about $PT_1$ and $F_2$ about $PT_2$ respectively, then

(A) $\angle PS_1F_2 = \angle PF_1S_2$ (B) $2\angle PS_1F_2 = \angle PF_1S_2$

(C) $4\angle PS_1F_2 = 3\angle PF_1S_2$ (D) none of these

82. Let differential equation of a curve f(x, y) = 0 is $\frac{dy}{dx}=\frac{-\left(xye^x\ln y+ye^y\right)}{xye^y\ln x+xe^x}$, and f(1, 1) = 0, then f(e, 1) is equal to

(A) e (B) $e^2$

(C) $\frac{1}{e}$ (D) none of these

83. Let $f(x) = 2x^3 - 9x^2 + 12x + 8$, the number of values of x for which y = f(f(x)) attains local extremum is/are

(A) 0 (B) 1

(C) 2 (D) 3

84. Let $\lim\limits_{x\to\pi}\frac{\sin\alpha x}{1-2\cos\beta x}=-\sqrt{3}$, $\alpha \in I^+$, $\beta > 0$, then least positive value of $\alpha\beta$ is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 1

***Space for rough work***

85. Let $\int \frac{\left(2x^2+3x-2\right)-\left(1+x^2-x\right)\ln\left(1+x^2-x\right)}{\left(x+2\right)^2\left(1+x^2-x\right)}dx = \frac{\lambda_1 \ln\left(1+x^2-x\right)}{2\lambda_2+\lambda_3 x}+\lambda_4 x + C$, then $\lambda_1 + \lambda_2 - \lambda_3 + \lambda_4$ is equal to

(A) 0 (B) 1
(C) 2 (D) 3

86. Let $y = m(x - 2) + 1$ and $y^2 = 4x$ bounds an area of A units, then the value of m if A is minimum is

(A) 1 (B) 2
(C) 3 (D) 4

87. Let $x^2 + \frac{1}{x^2} = 14$, then the value of $x^5 + \frac{1}{x^5}$ is equal to $(x > 0)$

(A) 70 (B) 700
(C) 724 (D) 770

88. The number of skew-symmetric matrices of order $4 \times 4$, whose entries belong to the set $\{-2, -1, 0, 1, 2\}$ is

(A) $5^9$ (B) $5^3$
(C) $3^6$ (D) $5^6$

89. The value of $\int_0^1 \left(2x^3 - 3x^2 + 3x - 1\right)^{\frac{1}{5}} dx$ is equal to

(A) $7\sqrt{5}$ (B) $5\sqrt{7}$
(C) $3\sqrt{5}$ (D) none of these

90. The line $x = 0$ divides the area enclosed by the curves $|x - 1| - y = 0$, $|x| + y - 3 = 0$ into two areas $A_1$ and $A_2$, $(A_1 < A_2)$, then $\frac{A_1}{A_2}$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

***Space for rough work***